250 results for “(survey OR review) AND (wildfire OR “wild fire” OR “wildland fire” OR “wild land fire” OR “forest fire” OR bushfire) AND (propagation OR spread OR behavior OR dynamics OR trajectory) AND (prediction OR forecast OR simulation OR model OR management)”

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Predicting the Occurrence of Forest Fire in the Central-South Region of China

Quansheng Hai | Xiufeng Han | +5 authors | Hailian Sun

Environmental Science, Geography · Forests · 11 May 2024

Understanding the spatial and temporal patterns of **forest fires**, along with the key factors influencing their occurrence, and accurately **forecasting** these events are crucial for effective **forest**… Expand

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An AutoML-Powered Analysis Framework for Forest Fire Forecasting: Adapting to Climate Change Dynamics

Shuo Zhang | Mengya Pan | Environmental Science, Computer Science · Atmosphere · 11 December 2024

TLDR This study provides an efficient **and** accurate method **for forest fire prediction**, which is of great significance **for** decision-making in **forest fire management and for** protecting **forest** resources **and** ecological security. Expand

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Integrating Multi-Source Remote Sensing Data for Forest Fire Risk Assessment

Xinzhu Liu | Change Zheng | Guangyu Wang | Fengjun Zhao | Ye Tian | Hongchen Li

Environmental Science · Forests · 18 November 2024

Forest fires are a frequent and destructive phenomenon in Southwestern China, posing significant threats to ecological systems and human lives and property. In response to the growing need for… Expand

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Exploiting Drone Images for Forest Fire Detection using Metaheuristics with Deep Learning Model

Environmental Science, Computer Science ·
Global NEST: the international Journal · 10 August 2023

TLDR This study develops an automated **Forest Fire** Detection using Metaheuristics with Deep Learning (FFDMDL-DI) **model** that exploits the DL concepts on drone images to identify the occurrence of **fire**.
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Simulating Forest Fire Spread with Cellular Automation Driven by a LSTM Based Speed Model

Xingdong Li Mingxian Zhang +4 authors Long Sun

Environmental Science, Computer Science · Fire · 20 January 2022

TLDR Based on the interaction between wind **and fire**, S-LSTM is proposed, which takes full advantage of the time dependency of the ROS **and** has good practicality in simulating the **spread** of **forest fires**.
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A Comparative Study on Forest Fire Prediction using ARIMA, SARIMA, LSTM, and GRU Methods

Arunkumar Gopu Anjali Ramakrishnan Ganesan Balasubramanian Kuna Srinidhi

Environmental Science, Computer Science ·
2023 IEEE International Conference on… · 21 April 2023

TLDR The results indicate that the LSTM **model** can effectively predict the occurrence of **forest fires** with high accuracy, with an MAE of 9.55 **and** an RMSE of 3.09, which can be used to develop early warning systems **for forest fires and** inform resource allocation **for** effective disaster **management**. Expand

4 IEEE Save Cite

Wildland Fire Spread Modeling Using Convolutional Neural Networks

J. Hodges B. Lattimer Environmental Science, Computer Science ·

Fire technology · 27 March 2019

TLDR The machine learning DCIGN approach was able to provide **fire spread predictions** at a computational cost three **orders** of magnitude less than current **models and** was shown to be capable of predicting

burn maps further in the future by recursively using previous **predictions** as inputs to the DCIGN. Expand

94 Springer Nature Save Cite

A comparison of three approaches for simulating fine-scale surface winds in support of wildland fire management: Part I. Model formulation and comparison against measurements

J. Forthofer B. Butler N. Wagenbrenner Environmental Science ·

13 November 2014 International Journal of Geographical Information Science

For this study three types of wind **models** have been defined for simulating surface wind flow in support of **wildland fire management**: (1) a uniform wind field (typically acquired from… Expand

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An Innovative Conceptual Model of a Forest Fire Management Information and Decision-Support System for Brandenburg State, Germany Abstract Research and development conducted within the Forest Fire…

Environmental Science, Engineering · 2016

Research and development conducted within the **Forest Fire** Cluster of the German Research Network on Natural Disasters is built on a number of separately evolved concepts that were integrated in a… Expand

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Real-time genetic spatial optimization to improve forest fire spread forecasting in high-performance computing environments

T. Artés A. Cencerrado A. Cortés T. Margalef

Environmental Science, Computer Science · International Journal of Geographical Information… · 1 March 2016

TLDR The Two-Stage calibration methodology, which has been shown to provide good results, is used **and** a new method which finds the minimum resolution reduction **for** such long **simulations**, keeping accuracy loss to a known interval is proposed. Expand

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